

“Auto Securitisation Credit Risk Analysis Dashboard Using Power BI”

Project Report Submitted for ZeTheta Internship Program

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Project Overview:

- The objective of this project is to analyze auto securitisation data and identify key credit risk indicators using Power BI. Auto securitisation is a financial process where vehicle loans are grouped together and converted into investment products. Monitoring the performance of these loans is important for understanding portfolio health and managing risk effectively.
- In this project, multiple datasets related to loan performance, customer information, default rates, and loss metrics were analyzed. The data was cleaned, transformed, and connected using Power BI to create an interactive dashboard. Various visualizations such as KPI cards, bar charts, line charts, and slicers were used to present the information in a meaningful way.
- The dashboard helps users monitor important metrics such as total defaults, monthly default rate, loss drivers, portfolio performance, and customer risk trends. The insights generated from the dashboard can support better decision-making and improve risk management strategies in the auto securitisation sector.

Dataset Description:

The Auto Securitisation Credit Risk Analysis project uses four datasets that contain information related to loan performance, customer details, defaults, and portfolio risk metrics. These datasets were integrated in Power BI to create a comprehensive dashboard for analysis.

- **Dataset 1: Loan Portfolio Data:**

This dataset contains information about the loans included in the securitised portfolio. It includes details such as loan amount, loan tenure, interest rate, outstanding balance, and loan status. This dataset was used to evaluate overall portfolio performance.

- **Dataset 2: Customer Information Data:**

This dataset contains customer-related information such as customer identification, demographic details, and other attributes associated with the loan accounts. It helps in understanding customer-level trends and risk patterns.

- **Dataset 3: Payment and Performance Data:**

This dataset contains repayment history and loan performance information. It includes records related to monthly payments, outstanding balances, and delinquency indicators. The data was used to analyze repayment behavior and identify potential risks.

- **Dataset 4: Default and Loss Analysis Data:**

This dataset focuses on defaults and loss-related metrics. It includes information such as default counts, monthly default rates, and loss drivers. The dataset was used to identify factors contributing to credit risk and assess portfolio stability.

Data Preparation and Data Cleaning:

Before creating the dashboard, the datasets were carefully prepared and transformed in Power BI to ensure accurate analysis and reporting. Data preparation is an important step because the quality of insights depends on the quality of the data being used.

The following data cleaning and transformation activities were performed:

- Checked the datasets for missing or blank values.
- Removed duplicate records wherever necessary.
- Verified and corrected data types for numerical and date fields.
- Created relationships between multiple datasets using common keys.
- Renamed columns for better readability and understanding.
- Applied filters to remove irrelevant or inconsistent records.
- Created calculated measures and KPIs using DAX functions.

After completing the data preparation process, the datasets were successfully integrated into a single data model. This enabled efficient analysis of loan performance, default trends, portfolio risk, and other key metrics through interactive Power BI visualizations.

The cleaned and transformed data provided a reliable foundation for building the Auto Securitisation Credit Risk Analysis Dashboard and generating meaningful business insights.

Dashboard Overview:

1.Auto Loan securitization Risk Analysis:

The Auto Loan securitization Risk Analysis provides a summary of the overall auto securitisation portfolio performance. It presents key metrics and visualizations that help users quickly understand the current status of the portfolio and identify important trends.

The dashboard includes various KPI cards and charts that display important information such as total loans, default counts, portfolio value, default rate, and other risk-related metrics. Interactive filters and slicers allow users to explore the data from different perspectives and perform detailed analysis.

The main objective of this dashboard page is to provide a high-level view of portfolio performance and support quick decision-making. Users can easily monitor portfolio health, track risk indicators, and identify areas that may require further investigation.

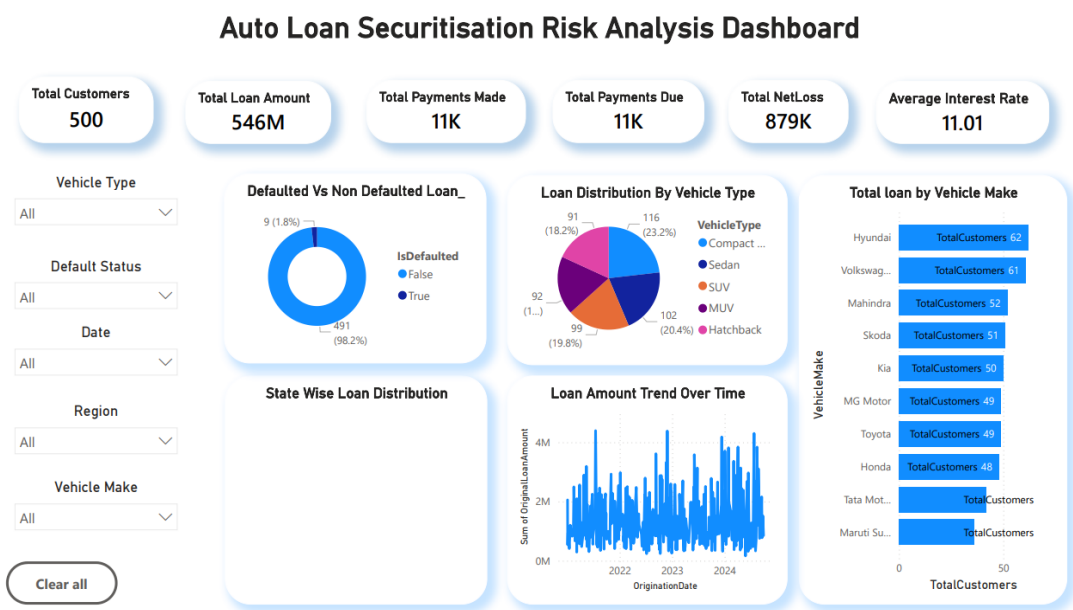


Figure 1: Auto Loan securitization Risk Analysis Dashboard

Key features of the dashboard include:

- Summary KPI cards for important business metrics.

- Interactive visualizations for trend analysis.
- Portfolio performance monitoring.
- Risk and default tracking.
- User-friendly filtering and drill-down capabilities.

This dashboard serves as the foundation for deeper analysis presented in the subsequent sections of the report and helps stakeholders gain a clear understanding of the overall portfolio condition.

2. Portfolio Performance & Collection Analysis:

This section focuses on the overall performance of the auto securitisation portfolio. The dashboard provides insights into loan distribution, portfolio value, repayment trends, and other important performance indicators.

The visualizations help in understanding how the portfolio is performing over time and whether the loan assets are generating stable results. By monitoring these metrics, stakeholders can evaluate portfolio strength and identify any areas that may require attention.

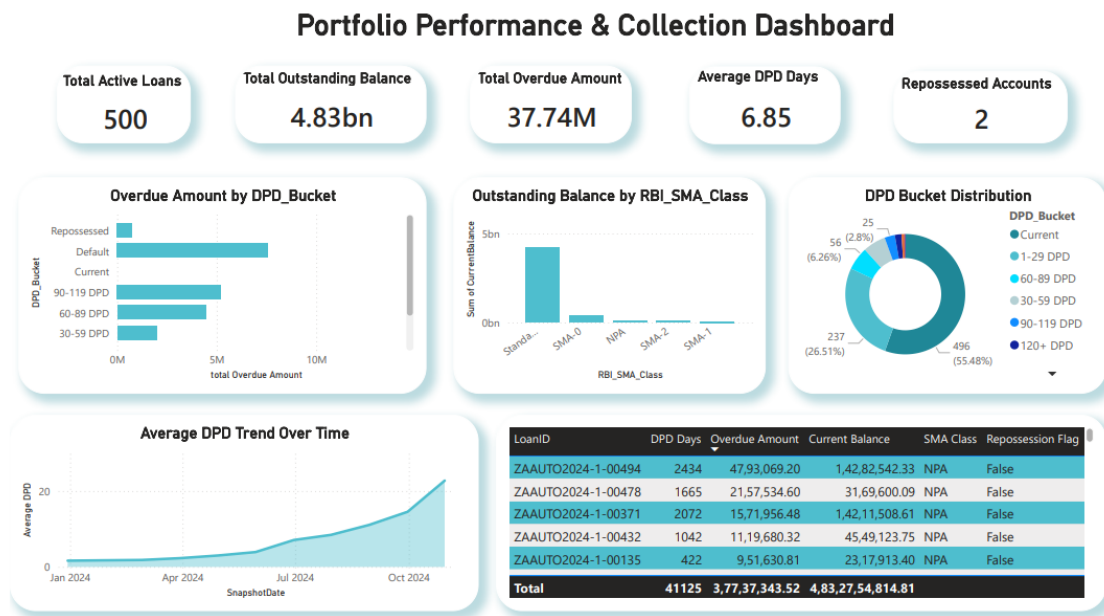


Figure 2: Portfolio Performance & Collection Analysis Dashboard

The dashboard allows users to analyze trends across different periods and compare performance indicators using interactive filters. This helps in identifying changes in loan behaviour and understanding the overall condition of the securitised assets.

Key observations from this dashboard include:

- Portfolio performance trends over time.
- Distribution of loans across different categories.
- Monitoring of key financial indicators.
- Identification of high-performing and low-performing segments.
- Better understanding of portfolio health and stability.

3. Loan Risk & Delinquency Analysis:

The Loan Risk & Delinquency Analysis dashboard focuses on identifying and monitoring loan defaults within the auto securitisation portfolio. Default analysis is an important part of credit risk management because it helps in understanding the level of risk associated with the loan portfolio.

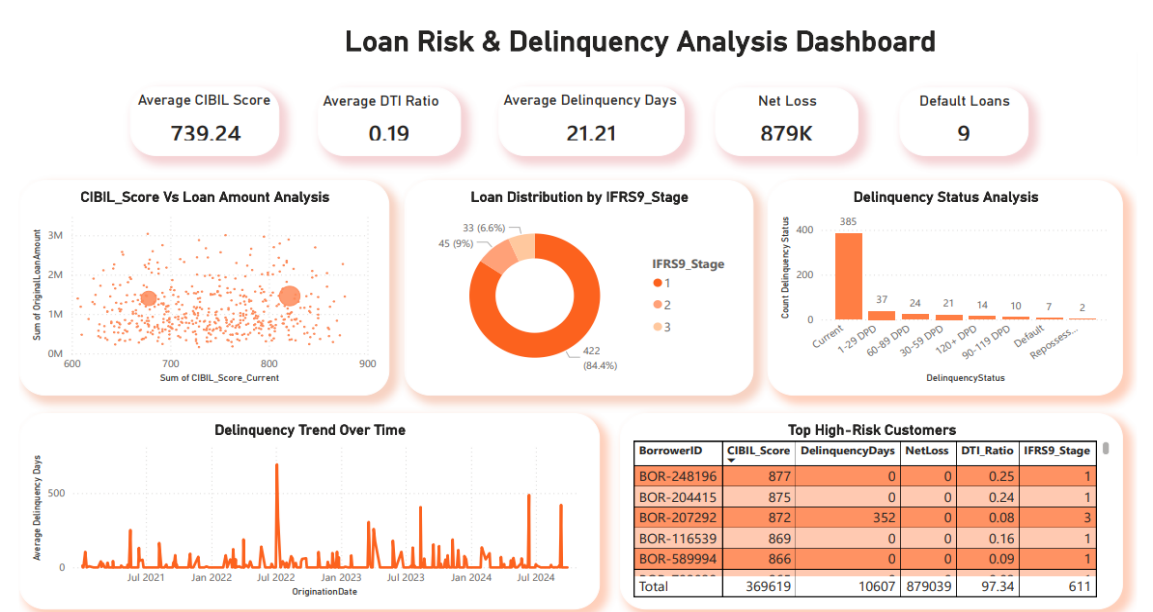


Figure 3: Loan Risk & Delinquency Analysis Dashboard

This dashboard provides insights into default counts, default rates, and trends over different time periods. The visualizations help users identify patterns in borrower behaviour and highlight areas where default risk is increasing.

By analyzing default-related metrics, stakeholders can assess portfolio quality and take proactive measures to reduce potential losses. The dashboard also helps in tracking the effectiveness of risk management strategies and monitoring changes in credit performance over time.

Key features of this dashboard include:

- Analysis of total default counts.
- Monitoring monthly default rate trends.
- Identification of high-risk segments.
- Comparison of default performance across different categories.
- Support for risk assessment and decision-making.

The insights generated from this dashboard can help financial institutions improve portfolio monitoring and strengthen credit risk management practices.

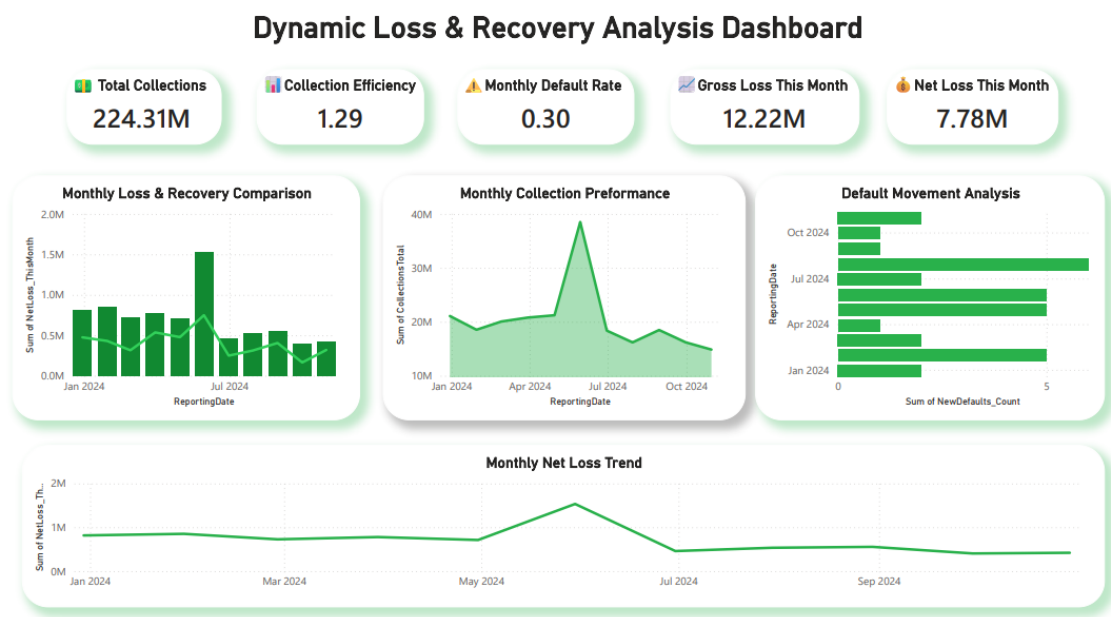
Dynamic Loss & Recovery Analysis:

The Dynamic Loss & Recovery Analysis dashboard focuses on identifying the major factors that contribute to losses within the auto securitisation portfolio. Understanding these loss drivers is essential for effective risk management and portfolio monitoring.

This dashboard helps users analyze the relationship between loan defaults and the factors that influence financial losses. By studying these patterns, stakeholders can identify potential risk areas and take corrective actions to improve portfolio performance.

The visualizations provide a clear view of how different variables impact losses and allow users to compare risk indicators across various segments of

the portfolio. This analysis supports better decision-making and helps financial institutions develop strategies to minimize future losses.



Key features of this dashboard include:

- Identification of major loss-driving factors.
- Analysis of the relationship between defaults and losses.
- Comparison of loss trends across different segments.
- Monitoring of portfolio risk indicators.
- Support for data-driven risk management decisions.

The insights obtained from this analysis can help improve portfolio quality, reduce credit risk exposure, and strengthen overall financial performance.

Vintage Pool performance Analysis:

The Vintage Pool performance Analysis dashboard is used to evaluate the performance of loan portfolios based on their origination period. Vintage analysis helps in understanding how different groups of loans perform over time and how risk evolves as loans mature.

This dashboard provides insights into the behaviour of various loan vintages and enables comparison between different origination periods. By analyzing vintage pools, stakeholders can identify trends in portfolio performance, repayment behaviour, and default patterns.

The visualizations help in tracking the performance of loan cohorts and understanding whether newer or older loan pools are exhibiting higher levels of risk. This analysis is valuable for monitoring portfolio quality and supporting long-term credit risk management.

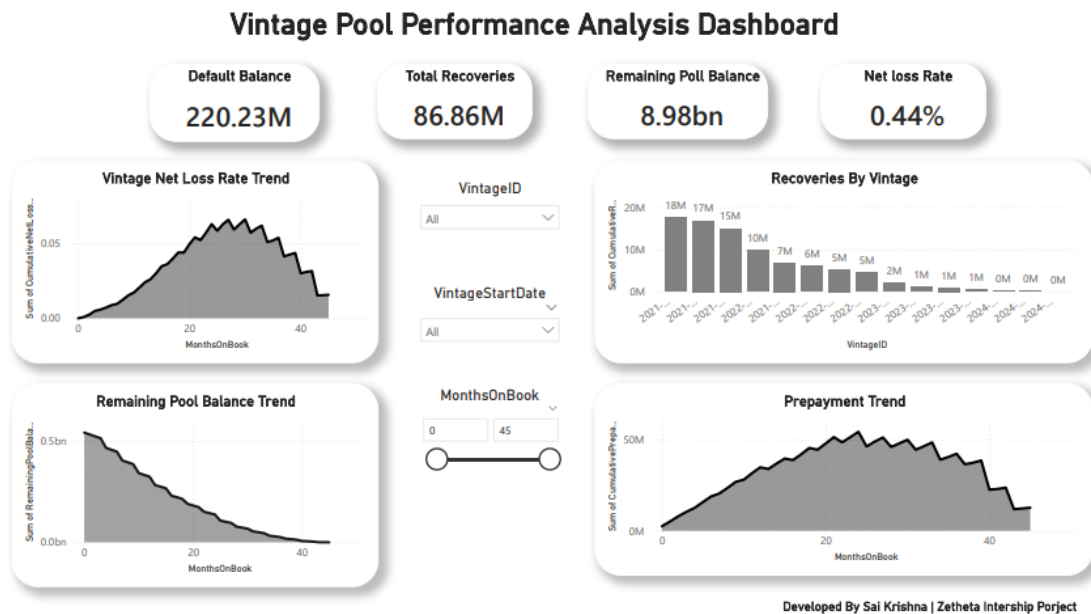


Figure 4: Vintage Pool Performance Analysis Dashboard

Key features of this dashboard include:

- Analysis of loan performance by vintage period.
- Comparison of different loan cohorts over time.
- Monitoring repayment and default trends.
- Identification of high-risk vintage pools.
- Support for portfolio monitoring and risk assessment.

The insights generated from vintage pool analysis help financial institutions better understand portfolio behaviour and make informed decisions regarding future lending and risk management strategies.

Key Insights and Findings:

Based on the analysis performed using the Power BI dashboard, several important insights were identified regarding the performance and risk profile of the auto securitisation portfolio.

Key Findings:

- The portfolio showed variations in performance across different time periods and loan segments.
- Default rates were not uniform and differed across customer groups and portfolio categories.
- Certain loan segments contributed more significantly to overall portfolio risk.
- Vintage pool analysis revealed differences in repayment behavior among loan cohorts originating in different periods.
- Monitoring default trends and vintage performance can help identify potential risks at an early stage.
- Interactive dashboard visualizations enabled quicker analysis and improved understanding of portfolio health.
- Data-driven insights can support better credit risk assessment and portfolio management decisions.

The dashboard successfully transformed raw data into meaningful information, allowing users to identify trends, monitor risk indicators, and evaluate portfolio performance efficiently.

Conclusion and Learning Outcomes:

Conclusion:

The Auto Securitisation Credit Risk Analysis Dashboard was developed using Power BI to analyze loan performance, default behavior, and portfolio risk. Through data cleaning, transformation, modeling, and visualization, the project provided valuable insights into the overall health of the securitised loan portfolio.

The dashboard enables effective monitoring of key performance indicators and supports informed decision-making by presenting data in an interactive and user-friendly format. The analysis highlights the importance of data analytics in credit risk management and portfolio monitoring.

Learning Outcomes:

During this project, the following skills and concepts were learned and applied:

- Data cleaning and transformation using Power BI.
- Data modelling and relationship creation.
- Development of DAX measures and KPIs.
- Creation of interactive dashboards and visualizations.
- Analysis of credit risk and default-related metrics.
- Portfolio performance and vintage pool analysis.
- Presentation of business insights through data analytics.

This project provided practical experience in using Power BI for financial data analysis and strengthened understanding of credit risk assessment and business intelligence techniques.